

Toxic Content Classification

Implementation report | 28 September 2026

The project combines LSTM classifiers trained in Task 1 with BLIP image captioning. A Streamlit interface accepts text, images, or both and records classification results in a CSV file.

Steps completed

- 1. Reused the trained models.** Loaded the query-only and combined-text LSTM models from Task 1, together with their saved tokenizers and settings in `task2_artifacts/`.
- 2. Added a prediction module.** `Task2_predict_text.py` classifies text alone or text plus a caption. Command-line checks confirmed that the saved models load and return a category and model score.
- 3. Added image captioning.** `imagecaption.py` uses the pretrained BLIP base model to turn an uploaded image into text. A local test with `sea.jpg` produced the caption `ocean waves.png`.
- 4. Built the Streamlit app.** `app.py` supports text-only, text-and-image, and image-only submissions. It displays the image caption, predicted category, and model score. Image-only prediction is preliminary because its classifier was trained on user text.
- 5. Added CSV history.** Every successful classification is appended to `classification_history.csv`. The app includes a **View saved history** section; a text-and-image submission was confirmed in the CSV.
- 6. Documented setup.** `requirements.txt` lists the Python dependencies, and `README.md` explains how to set up and run the project in VS Code.

Deployment

Platform	Streamlit Community Cloud
Repository	Abubaker-Nafe/cellula-tech-task-2
Branch / entry point	main / app.py
App address	https://cellula-tech-task-2.streamlit.app/

Local testing covered predictions, caption generation, the Streamlit interface, and CSV storage. Deployment is reported by the project owner; the supplied screenshot shows the deployment settings, and the public app was not independently checked for availability.